

INVITED REVIEW

Adaptive Introgression in the Context of Climate Adaptation

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ABSTRACT

As the biosphere faces accelerating environmental disruption including climate change and the prospect of an anthropogenically-driven mass extinction, understanding the mechanisms that enable species to adapt has become increasingly urgent. One mechanism attracting growing attention is adaptive introgression, the transfer of beneficial genetic variation between closely related species. Although frequently invoked as a potential driver of adaptation to rapid environmental change, including climate change, its overall importance remains debated. In this review, we adopt a balanced, evidence-based perspective. After introducing approaches for identifying potential donor and recipient species of introgression events, we review both the classical methods for detecting introgression and selection and the recent methodological advances that allow their joint inference. We discuss roughly 20 published studies representing taxonomic, geographic and climate diversity that provide at least partial support for a connection between introgression and climate-related adaptation, most often involving adaptation to climatic oscillations during the late Pleistocene rather than to contemporary climate adaptation. Although adaptive introgression is frequently invoked in evolutionary and conservation biology, we highlight a persistent gap between its theoretical expectation and the scarcity of well-documented, functionally or ecologically validated examples. Considering the pressing challenges to preserve biodiversity and improve predictions of adaptive potential under climate change, addressing this gap is of paramount importance. We conclude by proposing key research directions at the intersection of macroevolution, microevolution and the evolutionary ecology of communities.

1 | Introduction

One of the most pressing questions in evolutionary biology today is how species adapt to rapid environmental shifts, and whether such adaptive capacity will suffice to withstand the accelerating climate crisis. Adaptation is well known to be primarily driven by two mechanisms: new (de novo) mutations and pre-existing genetic variation within a population. The former is inherently constrained by the supply of new mutations, which is low in species with small contemporary population sizes (Bell 2013; Rousselle et al. 2020). The latter has long been overlooked (Hermisson and Pennings 2005, 2017;

Pritchard et al. 2010), but assembling new adaptive genetic combinations from standing genetic variation has recently become widely recognised as a key driver of rapid adaptation (Barrett and Schluter 2008). This is largely because standing variation of potential adaptive value is immediately available. Moreover, adaptation often has a polygenic basis, involving subtle shifts in allele frequencies across numerous loci and/or across regions of multiple small-effect loci in linkage disequilibrium rather than the fixation of single large-effect variants, allowing populations to respond more effectively to rapid environmental change (Barrett and Schluter 2008; Matuszewski et al. 2015). In this context, introgression—the transfer of

genetic material from distinct populations or closely related species through hybridisation followed by repeated backcrossing—is expected to enhance standing genetic diversity and potentially fuel adaptation through natural selection on introduced alleles that have evolved under different environmental conditions (e.g., Pfennig 2021; Taylor and Larson 2019). Our review focuses on the latter process, commonly referred to as adaptive introgression.

In this review, our objective is to synthesise the current knowledge about adaptive introgression in relation to climate, focusing on (i) the description of the process and its demographic underpinnings, (ii) the methodological approaches to detect the process, (iii) case studies in different climate contexts and (iv) several perspectives for further studies of climate-adaptive introgression and explorations of its possible relevance in a conservation context. To clarify the scope of our review, it is important to note two key points. First, by definition, introgression must precede adaptation in the recipient population or species. However, in most systems it is difficult to determine whether an introgressed allele confers an immediate advantage or whether selection acts on it only after a period of persistence in the standing genetic variation (see Zhang et al. 2021). In this review, we consider examples of adaptive introgression, without necessarily assuming that the adaptation occurred immediately after introgression. Second, we focus on non-domesticated species, since adaptive introgression in an agronomic context is already well documented (see the review by Burgarella et al. 2019 for plants, and also case studies in animals, e.g., Wang et al. 2020; Mary et al. 2022; Sarabia et al. 2025).

Before describing the process and synthesising the available evidence, it is important to clarify our position on the current controversy regarding adaptive introgression. The contribution of adaptive introgression to overall species adaptation remains a subject of debate and polarisation within the evolutionary biology community, with some researchers potentially overestimating and others dismissing its importance. Robust methods for quantifying the relative contribution of adaptive introgression to genomic adaptation are currently lacking, though some methods are currently under development (Jules Romieu, personal communication). This is because of the intrinsic challenge of disentangling patterns of adaptive introgression from classic sweeps while accounting for confounding genomic signals generated by incomplete lineage sorting, neutral introgression, background selection and polygenic adaptation (Moest et al. 2020; Zhang et al. 2020; Huang et al. 2025; Tourné and Chikhi 2025; Romieu et al. 2025). Collective efforts toward developing such methodologies are essential to move beyond the current polarisation and toward an objective, evidence-based understanding of the role of adaptive introgression across biota. In the absence of such standardised approaches, our objective here is to focus on the available empirical examples rather than to draw broad conclusions regarding the global prevalence. Adopting such a nuanced and cautious view is especially important when discussing adaptation to climate, given the current context of rapid global change and the projected risk of a sixth, anthropogenic mass extinction, at a time when we cannot afford to take a gamble on an overly optimistic assumption that widespread adaptive introgression will ensure climate resilience in most species.

2 | Neutral and Adaptive Introgression: Who, Where and When?

2.1 | Who: Introgression in Plants and Animals

A first question concerns the conditions under which species are able to exchange genetic material, or, more precisely, the level of divergence beyond which reproductive barriers preclude introgression. Such values can be critical in defining which species may act as potential donors or recipients of introgressible alleles. This question has received recent in-depth investigation through genome-wide demographic modelling across a broad range of plant and animal species (Roux et al. 2016; Monnet et al. 2025). This work has provided fundamental knowledge regarding dynamics of introgression across the continuum of divergence, providing threshold values (net divergence, d_A , of 0.3% in plants and 1.8% in animals, Monnet et al. 2025), although these values should not be considered too strictly. At similar levels of genetic divergence, plants appear to establish species barriers more rapidly than animals, consistent with more frequent reports of hybrid zones in animals than in plants (Abbott 2017; Nieto Feliner et al. 2023). Although botanists have long recognised the prevalence of hybridisation in plants, it has been differentially appreciated among authors in zoology (for historical perspectives, see Mallet 2007; Hedrick 2013; Suarez-Gonzalez et al. 2018 for instance; but see also Barton and Hewitt 1985; Mallet 2005), particularly among species pairs with more subtle morphological differences that could have previously been overlooked by zoologists, but that became apparent upon genetic examinations (e.g., Suzuki et al. 2005; Caputi et al. 2007).

Over the past two decades, human genomics has offered a new perspective by providing compelling evidence of archaic introgression between hominins such as Neanderthals and Denisovans (Green et al. 2010; Reich et al. 2010; Vernot and Akey 2015, among others). Increasing empirical evidence of neutral gene flow among closely related species, including plants, humans and other animals, suggests that introgression is a widespread phenomenon (Aguillon et al. 2022). This widespread occurrence likely reflects the fact that semi-isolated species pairs are more prevalent than traditionally assumed, highlighting a persistent semi-permeability of diverging genomes to gene flow during speciation (Harrison and Larson 2014). Following this view, rare gene flow events can introduce bursts of new genetic diversity and novel genetic combinations, even between well-diverged taxa in late stages of speciation. This outcome is especially likely when selective coefficients of adaptive loci are strong enough to surmount the effects of species barriers (Edelman and Mallet 2021) or alternatively, balancing selection facilitating the permeability of species barriers (Gaczorek et al. 2024), further raising questions about the relative contribution of introgression in adaptation.

2.2 | Where: Climate as a Driver of Range Shift and Introgression

Identifying potential donor and recipient species requires considering not only their level of divergence but also the spatial

context of the two species. Introgression is possible only when the ranges overlap or are close enough for dispersal to bridge the gap. Such situations give rise to demographic scenarios involving divergence with gene flow, or secondary contact that re-establishes genetic exchange between closely-related species. Past climate oscillations, especially the glacial–interglacial cycles of the last million years, have repeatedly triggered rapid range shifts across many taxa, driven by direct climatic changes, by the advance and retreat of continental ice sheets, and by indirect effects such as fluctuations in sea level. In Europe, for instance, the continent's highly heterogeneous geography has repeatedly generated fragmented ranges with small, isolated populations during glacial periods, followed by postglacial recolonisation and renewed contact among lineages during warmer periods (Hewitt 1999; Stewart et al. 2010). Consequently, such gene flow during glacial periods or following secondary contact in warmer periods has been documented in many European animal and plant species (e.g., in animals: flycatchers, Ellegren et al. 2012; honey bees, Leroy et al. 2024; in plants: oaks, Leroy et al. 2017; Leroy, Rougemont, et al. 2020; ashes, Heuertz et al. 2004, 2006; spruces, Zhou et al. 2024, among others). Even though geography plays a crucial role, for example, through geographical isolation during glacial periods, many life history strategies likely influence the occurrence of secondary contacts. Recently, Monnet et al. (2025) reported that secondary contacts are more frequently observed in plants than in animals for instance. In turn, this pattern likely contributes to the number of well-documented cases of adaptive introgression in plants.

2.3 | When: Adaptive Introgression Through the Holocene

The temporal resolution of introgression events can be achieved using recombination-based approaches that exploit the predictable breakdown of introgressed tracts over time (e.g., Pool and Nielsen 2009; Liang and Nielsen 2014; Duranton et al. 2018). In recent events, introgressed haplotypes remain relatively long and generate pronounced admixture-induced linkage disequilibrium, not yet degraded by recombination through time. Therefore, tract length and the decay of LD with genetic distance can be used to estimate the timing of gene flow. By contrast, ancient introgression events are characterised by extensive recombination, resulting in highly fragmented introgressed tracts and weak residual LD. This makes it much more challenging to detect introgressed segments and date gene flow. In introgressed genomes, regions under strong selection are detectable as long tracts at high frequencies, potentially confounded with recent introgression. However, regions under weak selection with subtle allele frequency shifts are difficult to detect, especially under ancient introgression.

Additional methodological constraints further bias empirical studies toward recent events. In older events, linking introgressed variants to traits is expected to be more difficult and the sampling of ancient or historical DNA remains technically demanding. These factors likely explain why the most compelling examples of non-human adaptive introgression come from contemporary or recent events (Taylor and Larson 2019), in contrast to the extensively studied archaic introgression in

hominins. Another factor that contributes to the higher number of recent adaptive introgression cases relates to the extent of human disturbance in the Holocene. Globalisation further contributes to species introductions and long-distance dispersal of species, including the spread of pathogens and pests, thereby promoting human-mediated introgression. In a survey of nature management practitioners across Europe, 160 cited cases of hybridisation (35%) involved an alien taxon (Myriam Heuertz, in preparation). A typical example involves marine species in major shipping ports, which serve as artificial environments where invasive species introduced via ship ballast water can hybridise with native species (e.g., Simon et al. 2020; Touchard et al. 2023). Even more strikingly, the widespread use of pesticides since the Second World War has imposed immense selective pressures on competitors in the environment, accounting for numerous well-documented cases of adaptive introgression in both animals and plants. In mice, alleles from *Mus spretus* have been introgressed into *Mus musculus* conferring resistance to rodenticides such as warfarin. This includes the genomic region of *Vkorc1*, a key warfarin resistance gene that enhances survival in *M. musculus* (Liu et al. 2015; Banker et al. 2022; see also Norris et al. 2015 for a similar example in insecticide-resistant mosquitoes). In plants, introgression of herbicide resistance alleles from agricultural crops into wild relatives has been reported for several species. For instance, introgression from cultivated maize into wild teosinte has introduced resistance alleles, allowing teosinte to persist in agroecosystems exposed to herbicides (Le Corre et al. 2020).

Beyond the direct anthropogenic impacts on current habitats and biodiversity, climate change is expected to shift many species distributions, increasing the likelihood of secondary contacts (Chunco 2014). Such new secondary contacts will occasionally facilitate adaptive introgression, potentially rescuing species by providing alleles that help them adapt to changing conditions (Jones et al. 2018; Razgour et al. 2019; Vedder et al. 2022; Brauer et al. 2023). However, well before discussing this point, the potential negative costs of such introgression, including the evolutionary impact of the elevated selection against foreign alleles due to deleterious load and hybrid incompatibilities, should also be considered (Harris and Nielsen 2016; Todesco et al. 2016; Adavoudi and Pilot 2021; Ramasamy et al. 2023; Leitwein et al. 2025).

3 | Methodological Approaches to Detect Adaptive Introgression

Demonstrating adaptive introgression requires clear evidence of introgression, that is, establishing that certain alleles originate from another species, and of adaptation, for example, genomic footprints of selection and correlation with a selective driver or evidence of fitness advantages. Consequently, adaptive introgression has often been demonstrated through a combination of methods, thereby generating independent support for both introgression and adaptation (Burgarella et al. 2019). We include a conceptual figure that shows an example stepwise methodological approach to assess evidence for adaptive introgression to cold climate comprising demonstration of 1. introgression, 2. selection, 3. functional relevance and 4. experimental validation (Figure 1), though deviations

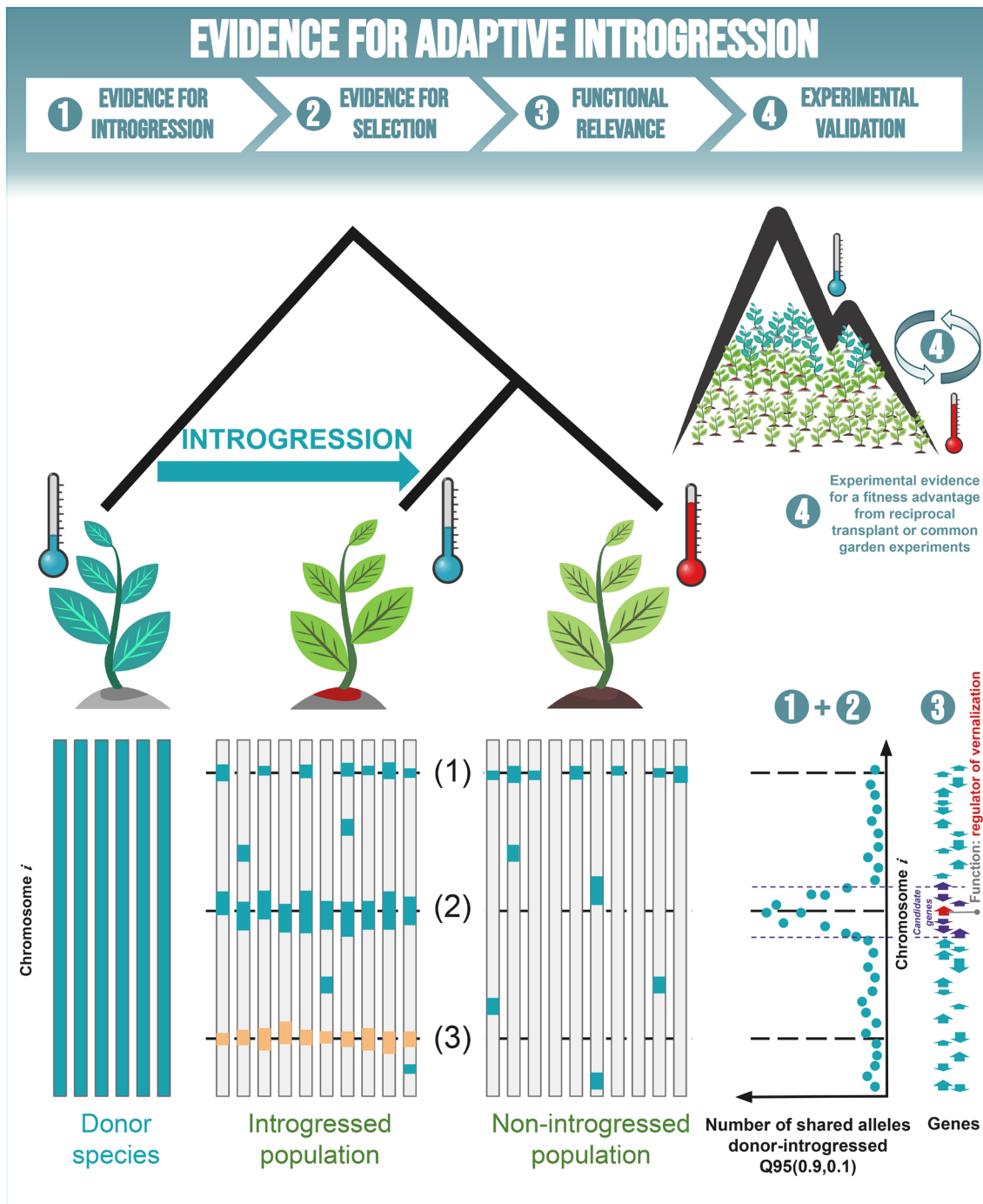


FIGURE 1 | Multiple sources of evidence for adaptive introgression through introgression, selection, gene function, and fitness effects as illustrated in a theoretical example of adaptive introgression contributing to cold tolerance of a plant species at high altitude. Joint inference of introgression and selection can also be performed, as illustrated using Racimo's Q95 summary statistic. Each vertical bar represents a copy of a given chromosome i , with blue segments indicating genomic regions carrying the same alleles as the donor population. The Q95 statistic (Racimo et al. 2017) is computed across the chromosome to detect signals of adaptive introgression, with values plotted along its length. Three illustrative cases are highlighted: (1) A region showing shared alleles in both introgressed (high elevation, i.e., lower temperature) and non-introgressed (low elevation, i.e., higher temperature) populations, resulting in no elevation of the Q95 summary statistic, potentially reflecting high incomplete lineage sorting (ILS) rather than selection. (2) A candidate region with a pronounced Q95 peak where donor alleles are at high frequency in the introgressed (high-altitude, i.e., cooler conditions) population, consistent with positive selection for cold-adaptive introgressed variants. (3) A region outside of introgressed segments displaying a potential hard selective sweep within the focal population (orange) rather than from introgression from the donor species (blue), resulting in no increase in the summary statistic. Although not all steps are strictly required, providing additional evidence on gene function (functional relevance) and fitness effects (experimental validation) substantially increases confidence in support of adaptive introgression.

from this approach are possible (see Section 4). Following a concise summary of historical approaches, we turn to recent developments and highlight the critical role of functional and experimental work.

3.1 | Methods to Detect Introgression

Although methods used for detecting introgression have been extensively reviewed already (e.g., Martin and Jiggins 2017; Suarez-Gonzalez et al. 2018; Burgarella et al. 2019), we provide a brief summary of key methods and how they enable differentiating a foreign ancestry from shared ancestry. A commonly used phylogenomic method is the D -statistic (commonly referred to as ABBA-BABA test) and its derivatives (e.g., Green et al. 2010; Malinsky et al. 2015; Martin et al. 2015). These methods are based on identifying local excess of allele sharing, beyond stochastic expectations under Incomplete Lineage Sorting (ILS). Whereas these methods are powerful in detecting genome-wide signatures of introgression, the D -statistic does not pinpoint specific introgressed regions, as its application to small genomic windows is affected by effective population size and high variability, especially in regions of low diversity or recombination, leading to potential false positives (Hibbins and Hahn 2022; Martin et al. 2015). Over the last decade, plenty of methods based on the conceptual framework of the D -statistic but overcoming its limitations in small genomic windows have been developed such as f_4 -tests (Green et al. 2010; Peter 2016), f_d (Martin et al. 2015) and f_{dM} (Malinsky et al. 2015). Other phylogenetically-oriented approaches such as the use of gene tree branch lengths, as implemented in QuIBL (Hibbins and Hahn 2022) or in Aphid (Galtier 2024), or topology weighting, as implemented in Twisst (Martin and Van Belleghem 2017) have also contributed to pinpoint local introgression footprints, alone or in complement with f_d and f_{dM} (Edelman et al. 2019; Ferreira et al. 2021; Yardeni et al. 2025, among others). In addition, several population genomics-oriented methods, including local ancestry inference tools like ELAI (Guan 2014), RFMix (Maples et al. 2013) or ChromoPainter-FineStructure (Lawson et al. 2012) and genome polarisation approaches like diem (Baird et al. 2023), are commonly used to identify introgressed tracts.

3.2 | Methods to Detect Selection

Although detecting introgression is crucial, it alone does not distinguish neutral from adaptive introgression. Indeed, detecting selection footprints in introgressed regions involves identifying genomic signatures that deviate from neutral expectations, therefore suggesting positive selection acting on the introgressed alleles. Most approaches are population genomics-oriented, either detecting selection by comparing patterns of genetic variation along the chromosomes within populations, or by looking at population differentiation. Within populations, a plethora of methods are based on different signatures of selection in genomic regions surrounding its target (i.e., selective sweeps), including locally reduced nucleotide diversity (π), skewed allele frequency spectra (excess of low- or high-frequency alleles, Tajima's D), extended haplotype homozygosity (e.g., iHS,

XP-EHH; Klassmann and Gautier 2022) or increased linkage disequilibrium (Alachiotis et al. 2012). Among populations, loci exhibiting elevated allele frequency differentiation between populations (e.g., F_{ST} or $X^T X$ outliers; Günther and Coop 2013) or genotype-environment association (GEA) methods (BayEnv, BayPass, LFMM, or RDA; Günther and Coop 2013; Gautier 2015; Frichot et al. 2013; Forester et al. 2018) that identify statistical associations between allele frequencies and ecological variables, suggest that introgressed variants contribute to local adaptation. Additionally, comparative genomics methods, such as the McDonald-Kreitman test and related approaches (McDonald and Kreitman 1991), complement these population genomic-oriented methods by contrasting patterns of polymorphism and divergence to infer positive selection in genes within the introgressed regions.

3.3 | Cline-Based Methods for Detection of Introgression and Selection

Considering the importance of sampling several introgressed and non-introgressed populations, an approach to jointly infer introgression and selection is the analysis of geographic clines. Geographic clines (e.g., Barton and Hewitt 1985) describe changes in allele frequencies across geographic space and can provide evidence for both introgression and selection, particularly in the context of reproductive barriers (steep clines), but also for adaptive introgression (shifted cline centers). However, in practice, hybrid zones often depart from simple clinal variation, forming complex mosaic structures that challenge purely geographic analyses. This has motivated approaches that examine introgression independently of spatial structure. Early studies by Szymura and Barton (1986) and Rieseberg et al. (1999) introduced the concept of evaluating locus-specific introgression relative to a genome-wide admixture background. This conceptual framework was later formalised into explicit genomic cline models that analyse allele-frequency variation along a genome-wide hybrid index rather than across geographic space, and has since been progressively refined through a series of methodological advances (Barton and Baird 1995; Lexer et al. 2007; Gompert and Buerkle 2009, 2011, 2012; Gompert and Buerkle 2010; Fitzpatrick 2013). Bayesian genomic cline methods allow the identification of loci exhibiting excess introgression relative to genome-wide average (negative β parameter estimates under the Bayesian genomic cline model), which can be consistent with adaptive introgression (Gompert and Buerkle 2012). This approach has for instance been used to identify candidate loci for adaptive introgression in a pine species from the eastern Tibetan Plateau (Guo et al. 2023) and in sand dune adaptation in the herbaceous *Armeria species* complex (Villa-Machío et al. 2023). These methods are now increasingly scalable to large genomic datasets (Gompert et al. 2024), providing an additional framework for genome-wide exploration of adaptive introgression patterns. Nevertheless, their results should be interpreted with caution and ideally confirmed using complementary approaches, as genomic clines do not provide unequivocal evidence for adaptive introgression but rather identify loci with introgression patterns that deviate from neutral genome-wide expectations.

3.4 | Advancing Methods for Jointly Detecting Introgression and Selection

Methodological developments have recently allowed shifting toward jointly detecting introgression and selection on introgressed loci. Compared to separate approaches, integrated analyses are expected to better capture subtle or overlapping signals and to mitigate confounding factors such as incomplete lineage sorting and complex demographic histories. To be applicable, most of these methods require sequence data from representatives of both the recipient and the donor species including introgressed and non-introgressed populations. Racimo et al. (2017) introduced several summary statistics, including the Racimo's Q95(w, y), a summary statistic specifically designed to identify regions with high-frequency derived alleles that are shared between the donor and recipient population, whereas these alleles are at low frequency in the non-introgressed sister population, providing a simple yet effective way to detect adaptive introgression (Figure 1). Contrary to the other methods, VolcanoFinder (Setter et al. 2020) does not require data from the donor, since it identifies 'introgression sweeps', a volcano-shaped pattern due to an excess of intermediate-frequency polymorphisms around the positively-selected site within the data from the recipient population alone (Setter et al. 2020). Empirical use of VolcanoFinder on Tibetan hominin genomes has revealed not only introgression at the *EPAS1* gene—a textbook locus of adaptive introgression (Huerta-Sánchez et al. 2014; Zhang et al. 2021)—, but also at other loci, highlighting the oligogenic nature of adaptation to hypoxia through archaic human introgression (Ferraretti et al. 2024).

Genomatnn (Gower et al. 2021) and MaLAdapt (Zhang et al. 2023) employ deep learning and machine learning approaches, respectively, to classify genomic windows as resulting from adaptive or non-adaptive introgression. Concretely, genomatnn uses deep learning with convolutional neural networks trained on simulated genotype matrices, specifically designed to distinguish adaptive introgression from confounding neutral processes and demography, as well as other forms of selection, in order to identify adaptive introgression on empirical data. MaLAdapt (Zhang et al. 2023) uses a machine learning approach that combines multiple statistics to distinguish adaptive from non-adaptive introgression. Although these tools were initially developed in the context of human evolution and have so far seen limited application to empirical data in other systems, recent simulation-based evaluations across diverse evolutionary scenarios suggest that their performance is highly sensitive to model assumptions and demographic contexts (Romieu et al. 2025). Notably, simpler approaches such as Racimo's Q95(w, y) summary statistic can perform comparably—or even outperform these learning-based methods—in terms of robustness and accuracy (Romieu et al. 2025). This outcome is particularly expected when machine learning-based methods are not sufficiently well trained, as their ability to learn adaptive introgression patterns strongly depends on the quality of the training data and its proximity to the testing set, especially with respect to underlying demographic history (Gower et al. 2021; Zhang et al. 2023; Romieu et al. 2025).

3.5 | The Crucial Importance of High-Quality Gene Annotations

An important, albeit often neglected, aspect is the integration of high-quality functional annotation. High-quality gene annotations are now accessible for a growing number of species (e.g., 276 annotations available in NCBI in 2015, 1406 available in 2025, a 409% increase), as well as through annotated metabolic pathways (e.g., +80 'KEGG organisms' species/month; Kanehisa et al. 2025) or expression data (e.g., from RNA-seq or ATAC-seq). All together, these approaches can help assess whether the candidate regions could indeed host genes that affect relevant traits. The favourable trend in terms of well-annotated species genomes is expected to accelerate over the next decade, driven by advances in gene annotation tools, including the emergence of deep learning-based strategies that have already started to increase the accuracy of gene prediction (Chen et al. 2024; Davison et al. 2025). Such high-quality investigations will likely surpass precise Gene Ontology (GO)-term enrichment analyses, which have become increasingly popular over the past decade or two for interpreting functional patterns in genomic data. GO-term analyses are, however, often affected by multiple sources of bias, including uneven annotation quality among species (experiment-based vs. bioinformatic annotations), overrepresentation of well-studied genes, and the use of overly broad or redundant terms (e.g., Gaudet and Dessimoz 2017; Strielkov et al. 2025). Major methodological flaws in enrichment testing have been reported in most studies, leading to widespread reproducibility issues (Yon Rhee et al. 2008; Wijesooriya et al. 2022).

Despite ongoing advances that solve some issues (e.g., Koopmans 2024; Valverde et al. 2025), some inherent challenges of enrichment analysis will, however, remain, including the fact that such approaches require making strong assumptions regarding the genetic basis of adaptation. For example, achieving significant enrichment in large gene families typically assumes a polygenic basis for adaptation, even though a single gene within a large family may still play a pivotal adaptive role. A similar caveat applies when GO-term enrichment analysis is conducted on all genes contained within a single chromosomal inversion, or in a recently introgressed large chromosomal region. Polymorphic inversions are often large, encompassing multiple genes, and segregate as single non-recombining blocks. As a result, the inversion behaves as a single unit and enrichment signals may simply reflect a difference in local gene content rather than a genuinely polygenic basis. In such large structural arrangements, linked selection makes it typically difficult to disentangle monogenic or oligogenic selection acting on one or a few causal genes within the inversion from true polygenic selection distributed across multiple genes located within the rearranged region. Consequently, while gene enrichment analysis can theoretically offer useful insights in the context of adaptive introgression, its outputs alone should be interpreted with caution.

3.6 | The Critical Need for Experimental Validation

Direct experimental evidence of enhanced fitness of introgressed versus non-introgressed individuals is the most compelling

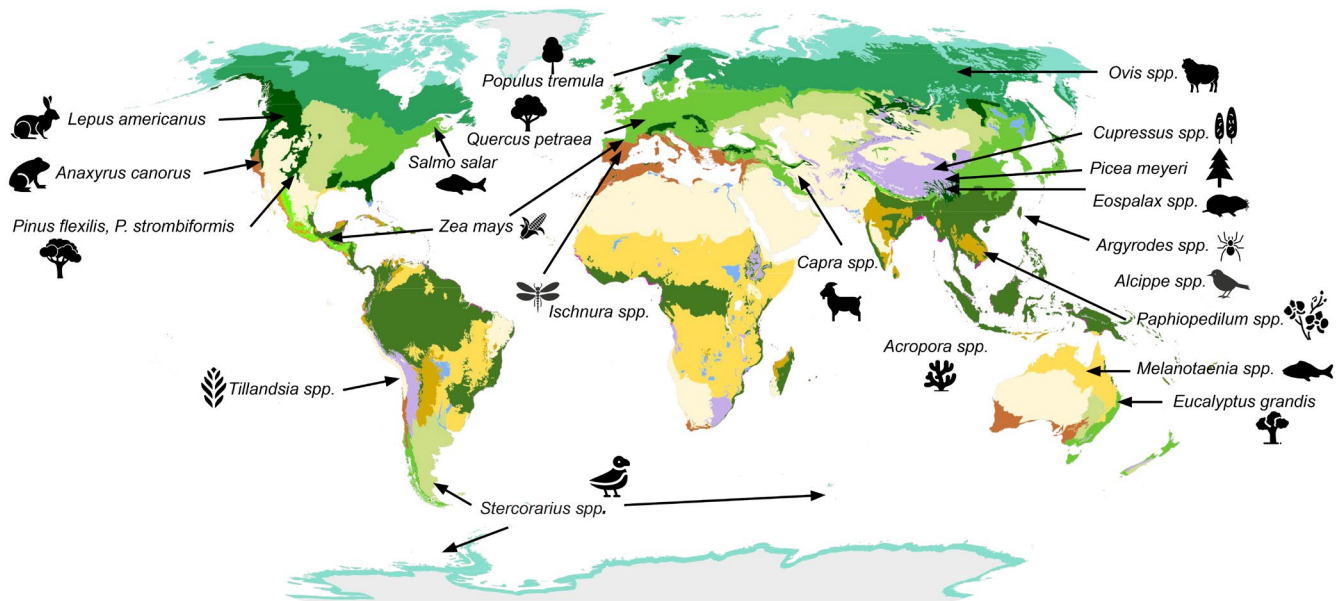


FIGURE 2 | Example cases of climate-relevant adaptive introgression described in the paper. The background map represents Terrestrial biomes of the world according to Olson and Dinerstein (1998) and used by the WWF and Global 200, downloaded from Wikipedia (CC BY-SA 3.0). Free icons were downloaded from www.flaticon.com.

support for adaptive introgression. Such investigations typically involve linking genetic variation to phenotypic traits and demonstrating that these traits improve performance—such as increased survival, reproduction, or stress tolerance—in the laboratory or in (semi-)natural environments such as in common garden or reciprocal transplant experiments. Textbook examples of adaptive introgression in the adaptation of trees to changed climatic conditions were at least partially based on such common garden strategies (e.g., Suarez-Gonzalez et al. 2016; Leroy, Louvet, et al. 2020; Rendón-Anaya et al. 2021). Powerful, field-based experiments, including common garden or reciprocal transplant studies, are often difficult to implement in practice, as they tend to be time-consuming, expensive, and particularly challenging for non-model, long-lived, or threatened species (Taylor and Larson 2019). Albeit crucial, such long-term experiments are further constrained by several factors, including the short-term grant funding system, which threatens continuity and limits the development of new experimental setups, as well as by additional concerns such as the potential biological risks associated with such transfers (e.g., conservation-related, long-distance pathogen dispersal and evolution, and so forth Leroy et al. 2016; Ramírez-Castañeda et al. 2022). In animals, where ethical aspects may limit experimental approaches, in natura observations can provide information on fitness differences, at least when adaptive introgression leads to a distinctive phenotype. An example is the snowshoe hare in which introgressed coat colour camouflage affects survival (Zimova et al. 2016).

4 | Case Studies of Adaptive Introgression in a Climate Context

To identify cases of adaptive introgression in relation to climate, we conducted a systematic search in Pubmed with the terms *adaptation AND introgression AND climate* yielding 15,318 studies published between 1957 and 2025, which when refined to (*adaptive introgression* [Body—All Words] AND *climate* [Body—All

Words] NOT *'crop'* [Body—All Words]) narrowed down to 353 studies published between 2007 and 2025 (search conducted on 12 November 2025). The results cover a broad range of taxa, life forms and life history strategies, as well as climate zones and climatic covariates. Climate parameters are widely hypothesised to act as potential drivers of introgression and to shape the associated species response traits. We selected example cases from the literature search aiming to cover taxonomic, geographical and climate diversity (Figure 2, Table 1), complemented by additional targeted searches on these examples, and present them in the text sorted according to their climatic covariates.

We applied the conceptual framework lined out in Figure 1, assessing the strength of the evidence for each of the steps, to which we add an assessment of the (cumulative) evidence for climate-driven adaptive introgression (Table 1). Methods for the latter were not explicitly included in Section 3 because it represents aggregated evidence gathered in methodological steps, strengthened by additional, case-study specific evidence that is more loosely connected to the methodological framework, for example, evidence for the association of a climate or climate-associated variable with a phenotype, evidence for a range expansion or niche shift, a demonstration of a past secondary contact, a temporal congruence of historical climate change with introgression events, etc. We also note that case studies varied in the amount of genetic or genomic information included, the number of samples and evolutionary lineages involved/studied, and the order and association of the different methodological lines of evidence assessed, so that fitting the studies into the table format was somewhat challenging.

This section aims to highlight common climatic drivers of adaptive introgression, but is probably not exhaustive. Other climate-relevant putative drivers exist, for example, in the marine environment, but are less well understood such as sea level

TABLE 1 | Selected cases of climate-relevant adaptive introgression described in the paper (section 4), along with an assessment of the strength of evidence for 1. Introgression (I), 2. Selection (S), 3. functional relevance (F) and 4. experimental validation (V), as well as 5. cumulative evidence (C) for climate-driven adaptive introgression.

Species/Taxa	Phylum/ life form	Genetic data/marker and sampling	Methods used for the detection of introgression	Methods used for the detection of selection	Methods used to assess functional evidence	Validation	Climatic co-variate and adaptive introgression signature	References	Strength of evidence ^a				
									I	S	F	V	C
<i>Acropora</i> sp.	Corals	Whole-genome sequences <i>n</i> = 5, one genome per species (<i>Acropora digitata</i> , <i>A. echinata</i> , <i>A. gemmifera</i> , <i>A. subglabra</i> , <i>A. tenuis</i>)	Phylonetwork approaches, D-statistic, speciation with isolation/migration modeling, cluster analysis for co- introgressed genes	dN/dS	Demography- climate association, GO terms analysis, literature searches	NA	Sea-level change (rise), faster evolution in introgressed genes preceding the main introgression event	Mao et al. (2018)	●	●	●	●	●
<i>Alcippe</i> spp.	Birds	Whole-genome sequences <i>n</i> = 78 across 3 species, <i>Alcippe davidi</i> , <i>A. fratercula</i> and <i>A. hueti</i>	ADMIXTURE, hybrid index, HyDe, D statistic, FILET	Selective sweep analysis, GEA with climate, genetic connectivity	Niche models, genomic offset vs. future climate, polygenic adaptation forward simulations, GO terms analysis, functional annotation, literature searches	NA	Warm-humid climate; introgression into temperate species associated with adaptation to warm-humid conditions and reduction of climate vulnerability	Zhang et al. (2025)	●	●	●	●	●
<i>Anaxyrus canorus</i>	Amphibian	Reduced representation (ddRAD-seq) <i>n</i> = 535 tadpoles mapped on transcriptome, 3 contact zones involving pairs of a total of 4 lineages	STRUCTURE, NewHybrids, HIEST, PCA, islands of divergence with dendropy and regions of admixture with genomic cline analysis (bgc)	Divergence outliers (highly diverged or highly admixed), GWAS with tadpole traits	GO terms enrichment and SNP effects (VEP) in islands of divergence vs. rivers of admixture, literature	NA	Desiccation; LPIN3 regulating tadpole growth during desiccation introgressed in 1/3 hybrid zones	Maier et al. (2024)	●	●	●	●	●
<i>Argyrodotes lanyuensis</i>	Spider	RAD-Seq, <i>n</i> = 130 from 15 localities on 9 islands	STRUCTURE, SNAPP, ASAP species delimitation, hybrid index, D statistics, DIYABC-RF demographic history	Natural selection by comparative population genomic analysis, CEGA	Blastx annotation, literature searches	NA	Tropical monsoon climate; hybrid lineage and balancing selection on stress- relevant gene on Green Island with intermediate climate (tropical/ subtropical)	Response et al. (2025)	●	●	●	●	●

(Continues)

TABLE 1 | (Continued)

Species/Taxa	Phylum/ life form	Genetic data/markers and sampling	Methods used for the detection of introgression	Methods used for the detection of selection	Methods used to assess functional evidence	Validation	Climatic co-variant and adaptive introgression signature	Strength of evidence ^a				
								References	I	S	F	V
<i>Capra</i> sp.	Mammalia	Whole-genome sequences $n = 50$ ancient, $n = 211$ domestic, and $n = 72$ wild <i>Capra</i> genomes	ADMIXTURE, fineSTRUCTURE, ADMIXTOOLS and Dsuite (outgroup-f3, D, F-branch, F4- ratio), Treemix	Selective sweep & FST outlier analysis	Dating admixture event (SMC+), annotation based on reference genome, literature searches, historical samples carrying KITLG region under selection	KITLG acts on melanoblasts/ melanocytes, association with coat colour, environment and "domestication", stress tolerance and litter size (An et al. 2012); review on role in adaptation (Lu et al. 2025)	Adaptation to drought; introgression of hair-follicle adaptation from Nubian ibex into Levant goats coincident with wet-to-dry transition	Asadollahpour et al. (2023); experimental: An et al. (2012)	●	●	●	●
									●	●	●	●
<i>Cupressus gigantea/C. duclouxiana</i>	Trees	Transcriptome (RNA-Seq) $n = 30$ <i>Cupressus duclouxiana</i> , $n = 35$ <i>C. gigantea</i> trees	Frappe, coalescent simulations, Identity- by-descent analysis, D, fdM, dxy statistics	HKA test, shared loci under selection	GO terms enrichment, literature searches, past ranges through niche modelling	NA	Adaptation to high altitude of northern <i>C. duclouxiana</i> , carrying introgressed genes under selection, e.g., for light and general stress response	Ma et al. (2019)	●	●	●	●
									●	●	●	●
<i>Eospalax baileyi, E. cansus</i>	Mammalia	Whole-genome sequences (low coverage), $n = 230$ individuals from 19 populations, high and low-altitude species	ADMIXTURE, D and fd statistics, coalescent simulations, TreeMix	Shared loci with selective sweeps (low π , high FST)	GO terms and KEGG pathways enrichment, literature searches	NA	Adaptation to high altitude, adaptation to hypoxia facilitated by introgression of cardiovascular and hypoxia- related genes	Kang et al. (2024)	●	●	●	●
									●	●	●	●
<i>Eucalyptus grandis</i>	Trees	SNP array 60K $n = 362$ individuals from 33 provenances, and five outgroup species with $n = 10$ -18/species	sNMF (LEA package), Efficient Inference of Local Ancestry (EILA) R package	Environmental association with LFMM, GDM, LD patterns for detection of structural variants	GO terms enrichment, literature searches	NA	Adaptation to drier climates, with introgressed provenances occurring at lower dry period precipitation	Mostert-O'Neill et al. (2021)	●	●	●	●
									●	●	●	●

(Continues)

TABLE 1 | (Continued)

Species/Taxa	Phylum/ life form	Genetic data/marker and sampling	Methods used for the detection of introgression	Methods used for the detection of selection	Methods used to assess functional evidence	Validation	Climatic co-variant and adaptive introgression signature	References	Strength of evidence ^a				
									I	S	F	V	C
<i>Ischnura elegans</i> , <i>Ischnura graellsii</i>	Insects	Genetic marker datasets (SSRs), $n = 29$ <i>I. graellsii</i> , $n = 208$ <i>I. elegans</i> , from 14 populations	STRUCTURE ancestry proportions compared with those of simulated hybrids; morphology	NA	Ecological niche overlap of introgressed vs. pure populations	NA	Adaptation to warm temperate climate, niche expansion in introgressed populations	Wellenreuther et al. (2018)	●	●	●	●	●
<i>Lepus americanus</i> /L. <i>californicus</i>	Mammalia	Whole-genome sequences $n = 80$ from 4 US regions	PhylonetHMM, simulation of hybridization pulse with SELAM	Distribution of fitness effects, length of introgressed regions, inference of TMRCA (for the Agouti region)	Annotation based on reference genome, literature searches, previous work	Animals mismatched with colour background have up to 7% lower weakly survival, Zimova et al. 2016	Expansion into mild climate range of L. americanus, introgression of winter-brown camouflage (Agouti)	Jones et al. (2020); experimental: Zimova et al. (2016)	●	●	●	●	●
<i>Melanotaenia</i> spp.	Fish	ddRAD-Seq $n = 344$ fish from five species	ADMIXTURE, hybrid index, NewHybrids, D, f_4 -ratios and fdM statistics (Dsuite), TreeMix	GEA (RDA) with climate, overlap of GEA and introgressed genes with fdm	GO terms enrichment, literature searches, past and future range and genomic vulnerability with niche models, genomic offset	Discussion mentions consistency with evidence from experiments of adaptive resilience to projected climates, providing references	Adaptation to climate warming in altitudinal/ temperature gradient, with lower vulnerability (niche loss) of introgressed populations	Brauer et al. (2023), experimental: McCairns et al. (2016); and references in Brauer et al. (2023)	●	●	●	●	●
<i>Ovis nivicola</i> , <i>Ovis</i> sp.	Mammalia	Whole-genome sequences $n = 6$ <i>O. nivicola</i> , $n = 1$ <i>O.</i> <i>dallii</i> , $n = 1$ <i>O. ammon</i>	F4 statistics, TREEMIX, relative node depth (RND), Fst-based tree, DFOIL	Branch-site model in CODEML, Fay & Wu's H test for positive selection	Annotation based on reference genome, literature-based function	NA	Cold adaptation (to subarctic climate); introgression of immunity, adipogenesis and morphology- related genes	Upadhyay et al. (2021)	●	●	●	●	●

(Continues)

TABLE 1 | (Continued)

Species/Taxa	Phylum/ life form	Genetic data/markers and sampling	Methods used for the detection of introgression	Methods used for the detection of selection	Methods used to assess functional evidence	Validation	Climatic co-variate and adaptive introgression signature	Strength of evidence ^a				
								I	S	F	V	C
<i>Paphiopedilum wenshanense</i> , <i>P. concolor</i>	Orchid	RAD-Seq, <i>n</i> = 81 <i>Paphiopedilum wenshanense</i> , <i>n</i> = 9 <i>P. concolor</i> , <i>n</i> = 10 <i>P. bellatum</i>	neighborNet, ABC demographic modelling with fastsimcoal2, STRUCTURE, gene flow estimated with divMigrate	Environmental and flower morphology correlates with introgressed regions	species distribution modelling for temporal range shifts, niche similarity test	NA	Adaptation to warm, moist climate; higher introgression in more southern <i>P. wenshanense</i> is associated with warmer and moister climates	●	●	●	●	●
								Jiang et al. (2025)				
<i>Picea asperata</i> , <i>P. crassifolia</i> , <i>P. meyeri</i>	Trees	RNA-Seq, <i>n</i> = 114 from 21 populations of 3 species	ADMIXTURE, D-suite, TREEMIX, f _{DM}	Selective sweep analysis, GO-terms enrichment, environment-associated introgressed regions (RDA), niche differentiation with distribution modelling	Functional annotation, GO terms enrichment, literature searches	NA	Bidirectional adaptive introgression between allopatric species pairs affecting genes linked to stress resilience and flowering time.	●	●	●	●	●
								Feng et al. (2025)				
<i>Pinus flexilis</i> , <i>P. strobiformis</i>	Trees	ddRAD-Seq, 1122 trees from 132 pops	STRUCTURE, INTROGRESS	Gene-environment association (Bayenv2), RDA, LD network/ variance partitioning (SNP clusters)	Functional annotation, RDA	NA	Adaptation to freeze-related gradients enabled by introgression of SNP clusters from <i>P. flexilis</i>	●	●	●	●	●
								Menon et al. (2021)				
<i>Populus tremula</i> lineages	Trees	Whole-genome sequences <i>n</i> = 411 from 7 countries	NGSAdmix, EIGMIX (SNPRelate), D, f _d and f _{DM} statistics, df (PopGenome), ELAI, demographic simulations	iSAFE (integrated selection of alleles favored by evolution) for positive, betascan for balancing selection	GO enrichment based on reference genome, literature searches	Common gardens at high and low latitude in Sweden, high-latitude provenance with FT north-adapted allele does not grow in the south	Adaptation to short growing season at high latitude facilitated by introgression of flowering locus T adapted allele	●	●	●	●	●
								Rendón-Anaya et al. (2021), experimental: Michelson et al. (2018)				

(Continues)

TABLE 1 | (Continued)

Species/Taxa	Phylum/ life form	Genetic data/marker and sampling	Methods used for the detection of introgression	Methods used for the detection of selection	Methods used to assess functional evidence	Validation	Climatic co-variant and adaptive introgression signature	References	Strength of evidence ^a				
									I	S	F	V	C
Quercus petraea	Trees	Whole-genome poolseq (1 pool/pop) for 18 populations, 1 pop each of Q. petraea, Q. robur, Q. pyrenaica, Q. pubescens	TREEMIX, f3 statistic	GEA (Baypass) outlier identification (temperature, precipitation) and GPA (leaf unfolding date)	Manual annotation, literature searches, ecological clines vs. genomic ancestry cline association	Not directly, but complementary evidence for local adaptation from common gardens	Latitudinal and altitudinal gradients, higher introgression from Q. robur in cooler/ wetter locations	Leroy et al. (2020b)	●	●	●	●	●
Salmo salar	Fish	Genetic marker datasets (SNP array, 220k SNPs), n = 297 individuals from 11 populations	based on literature/ previous studies	pRDA, LFMM for genotype- phenotype association	GO terms enrichment, literature searches, RDA for GEA, trait- specific genomic offset analysis	NA	Climate warming, timing of long- distance marine- to-river migration, trait-specific genomic offset	Beck et al. (2025)	●	●	●	●	●
Stercorarius spp.	Birds	Whole-genome sequences n = 111 individuals from 21 locations, 3 species	ADMIXTURE, D and f4 statistics, TREEMIX	Sweep Detection (RAiSD), LD patterns (XP- nSL), shared loci under selection	GO and pathways enrichment, future range prediction by niche modelling	NA	Adaptation to extreme environments through introgressed genes affecting thermoregulation facilitates polar colonisation	Jorquera et al. (2025)	●	●	●	●	●
Tillandsia spp.	Herbaceous plants	Reduced representation (GbS, plastome) n = 489 individuals from 3 species + 1 outgroup	STRUCTURE, TREEMIX, morphometrics, plastome haplotypes	niche modelling for spatiotemporal range overlap and niche differentiation	NA	NA	Adaptation to extreme aridity facilitated by hybridization and introgression	Stein et al. (2023)	●	●	●	●	●
Zea mays (teosinte)	Herbaceous plants	50k SNParray, n = 70 accessions French teosintes, n = 40 Spanish teosintes, n = 314 acc. Zea parviglumis, n = 332 acc. Z. mexicana, n = 94 Z. mays landraces, n = 155 Z. mays inbred lines	FastStructure, F4 statistics, TREEMIX, ELAI	Differentiation scan (pcadapt), patterns of Fst and LD	Annotation based on reference genome, literature searches	Common garden experiment: male flowering adapted to French photoperiod in European introgressed but not Mexican teosintes	Adaptation of flowering time to temperate climate (shorter in introgressed temperate- adapted teosinte)	Le Corre et al. (2020)	●	●	●	●	●

^aStrength of evidence: the column headers stand for I, introgression; S, selection; F, functional relevance; V, validation; C, cumulative evidence for climate-driven introgression. Grey circle, not assessed or no evidence; light green circle, weak evidence; dark green circle, strong evidence.

change resulting in adaptive introgression among *Acropora* coral species (Mao et al. 2018). The cited examples illustrate frequent co-introgression of adaptation-relevant genes, for example temperature, precipitation and phenology-related genes (Feng et al. 2025; Leroy, Louvet, et al. 2020).

4.1 | Adaptation to Drought or Dry Environments

Adaptation to drought or dry environments is a frequent requirement under climate change. In the toad *Anaxyrus canorus*, a genomic analysis of introgression patterns and a GWAS analysis with tadpole development traits identified 10 genomic locations including LPIN3, a lipid metabolism gene, potentially involved in desiccation tolerance. This gene was present in genomic regions of high among-lineage divergence across three contact zones, but was affected by introgression at one of these contact zones (Maier et al. 2024). In mammals, adaptation to drought involves temperature regulation that is tightly linked to skin and hair traits. This is exemplified in indigenous goats in Southwest Asia adapted to desert climate, in which the KITLG region affecting hair follicles and related to heat loss capacity was found to be introgressed. The possible origin of this drought-adaptive KITLG region lies in the Nubian ibex from which it introgressed into domestic goats, and subsequently into Southwest Asian wild animals (Asadollahpour Nanaei et al. 2023).

In plants, there are also examples of introgression in relation to drought adaptation, especially in forest trees. In *Eucalyptus grandis*, provenances showed extensive interspecific introgression in genomic regions enriched for defence response and signalling genes in response to increased aridity (Mostert-O'Neill et al. 2021). In a complex of three allopatric spruce species in China, introgression of root-morphology and stress tolerance genes from the drought-adapted *Picea crassifolia* into the threatened *P. meyeri* likely enhanced the environmental niche and ecological plasticity of the latter (Feng et al. 2025). Climate-driven interspecific gene flow associated with hyperarid desert habitats is also exemplified by plants of the genus *Tillandsia*, in which introgressed genotypes increase adaptive diversity maintained through clonal reproduction in highly stressful environments where sexual reproduction is extremely rare (Stein et al. 2023).

4.2 | Adaptation to Cold Climate Gradients

Adaptation to cold environments has been facilitated by adaptive introgression in both plants and animals. Among mammals, the snow sheep *Ovis nivicola*, a wild herbivore inhabiting cold and alpine habitats of northeastern Siberia, shows evidence of introgression with the high-altitude argali sheep (*Ovis ammon*) and Dall sheep (*Ovis dalli*), likely resulting from contact during the Pleistocene. The introgressed genomic segments include genes associated with immunity, adipogenesis and morphology-related traits, representing potential targets of adaptive introgression. Notably, adipogenesis-related genes displayed introgression across several species boundaries in sheep and were under selection in snow sheep (Upadhyay et al. 2021). Another interesting case of adaptation to cold,

high altitude environments concerns Myospalacinae, a group of subterranean rodents. In this group, low-altitude Gansu zokors (*Eospalax cansus*) have introgressed cardiovascular-related genes associated with hypoxia adaptation from the high-altitude plateau zokors (*Eospalax baileyi*), which likely facilitated their adaptation to higher-altitude grass-land ecosystems of the Qinghai-Tibet Plateau (Kang et al. 2024). In bird species of the genus *Stercorarius*, colonisation of polar environments in the southern hemisphere was facilitated by hybridisation, with signals of selection in hybrids for genes involved in thermoregulation, for example, energy and fatty-acid metabolism (Jorquera et al. 2025).

Several examples have also been observed in trees. In an extensive hybrid zone between *Pinus strobiformis* and *P. flexilis* across western North America, introgressed variants from the cold-tolerant *P. flexilis* were preferentially retained along freeze-susceptible environmental gradients, whereas background variants were favoured along gradients related to water availability (Menon et al. 2021). Adaptive introgression also likely enabled the cypress *Cupressus duclouxiana* to expand its range to higher altitudes on the Qinghai-Tibet Plateau. Indeed, northern populations of *C. duclouxiana* exhibit 16 genes under selection, enriched for stress tolerance functions, that introgressed from the related high-altitude *C. gigantea* (Ma et al. 2019). In European white oaks, *Quercus petraea* populations from cooler environments along latitudinal and altitudinal gradients exhibit a higher level of introgression from *Q. robur*. Genomic regions introgressed from *Q. robur* involved in adaptation to these environments were associated with both water stress and phenology (leaf-unfolding in common gardens) and include several genes controlling stomatal responses such as RopGEF1 and PBL10 (Leroy, Louvet, et al. 2020). Adaptive introgression in oaks is not limited to European species but has also been documented in Asian species (Fu et al. 2022).

4.3 | Adaptation to Climate Warming

Introgression conferring adaptation to climate warming is exemplified in species that expand their ranges from cold or harsh climates into milder areas. An interesting case is the snowshoe hare (*Lepus americanus*), widely distributed in the boreal climate zone in North America. Although the species typically develops white winter fur, some populations exhibit brown winter camouflage due to a selective sweep at the Agouti pigmentation gene, which is introgressed from black-tailed jackrabbits (*L. californicus*; Jones et al. 2018). Although a hybridisation pulse between these species has been dated to the end of the last glacial maximum, the selective sweep was probably more recent and facilitated southward range expansion into areas with a milder climate (Jones et al. 2020). In damselflies, introgression from the Iberian *Ischnura graellsii* enabled niche expansion of *I. elegans* and colonisation from France into the Iberian peninsula (Wellenreuther et al. 2018). Introgression of warm-advantageous traits may reduce the carrier population's vulnerability to contemporary climate warming. Demonstrating this process is delicate but has been done in an altitudinal gradient in the *Melanotaenia* rainbowfish species complex in the Australian Wet Tropics in which high-altitude narrow endemic lineages hybridising with a widespread low-altitude congener had reduced vulnerability

to projected climates compared with pure narrow endemics. The hybrid lineages displayed introgressed regions enriched in genes with regulatory functions on heat shock proteins and cellular responses to thermal stress (Brauer et al. 2023). In the Sino-Himalayan fulvetta bird species complex, introgression of alleles linked to warm-humid tolerance from *Alcippe hueti* into the temperate-adapted higher-altitude *A. davidii*, for example, KBTBD2 mediating cellular response to heat stress, was associated with reduced climate change vulnerability in the latter species (Zhang et al. 2025). Introgression during range expansion conferring adaptation to warm climates can also be detected as balancing selection, which is the case in the kleptoparasitic spider *Argyroides lanyuensis* endemic to the area between subtropical Taiwan and the wet tropical Philippines, in which the climatically intermediate Green island was colonised by a hybrid lineage featuring stress-tolerance genes under balancing selection (Responte et al. 2025). A plant example concerns the southward range expansion of the orchid *Paphiopedilum wenshanense* in which introgression from *P. concolor* occurred preferentially in the southernmost populations and was associated with climatic covariates such as increased rainfall seasonality and annual mean temperature (Jiang et al. 2025).

4.4 | Climate Adaptation Through Phenology

Adaptation of phenology associated with climatic conditions frequently becomes necessary as a consequence of migrating to new latitudes. For example, in migrating fish species such as Atlantic salmon, run time, that is, the timing of return to natal rivers for spawning, is a highly fitness-relevant trait. In North American Atlantic salmon, both early and late run time were associated with the transatlantic introgression of a large structural variant, maturation-associated genes previously characterised in Atlantic salmon, and *ppia2*, a migration-timing gene conserved across vertebrates (Beck et al. 2025). In plants, a typical example is the European aspen, *Populus tremula*, in which northern Scandinavian populations are homozygous for a ~500 kb region on chromosome 10 introgressed from more eastern populations and carrying two *Flowering Locus T* homologues; these are photoperiod-relevant genes that contribute to adaptation to the short growing seasons characteristic of Northern Scandinavia (Rendón-Anaya et al. 2021). A similar case of phenology-relevant introgression conferring adaptation to high latitude climates concerns weedy teosintes found in maize fields of France and Spain. These weeds were found to originate from *Zea mays* ssp. *mexicana* race 'Chalco', a weedy teosinte from the Mexican highlands, and carried ZCN8, a major flowering time gene introgressed from the high latitude Dent maize grown in Europe, along with other introgressed regions conferring herbicide resistance (Le Corre et al. 2020).

5 | Perspectives Regarding Adaptive Introgression

5.1 | Past Adaptive Introgression and Diversification

Adaptive introgression is one of the evolutionary biology topics that lies at the interface between micro- and macroevolution. The methods we reviewed are typically oriented toward either

population genomics or phylogenomics, reflecting a conceptual and methodological division in the community that can limit our understanding of adaptive introgression. Bridging micro- and macroevolutionary perspectives is an important step toward understanding how adaptive introgression contributes to species diversification. Comparative genomic studies investigating large sets of species pairs along a continuum of divergence are increasingly common (e.g., Stankowski et al. 2019; Shang et al. 2023; Postel et al. 2025; Schield et al. 2025), providing a robust framework to quantify how reproductive isolation accumulates across lineages. Building on such large-scale comparative datasets, future research could integrate population-level analyses of adaptive introgression to identify when and how introgressed alleles persist or vanish across different stages of divergence. Developing this integrative approach, by combining population genomics, functional, ecological and phylogenomics, will help trace the fate of introgressed alleles from local adaptation to long-term lineage diversification. Fine-scale demographic inference, ecological context, paleogenomics and experimental validation can clarify the selective mechanisms underlying introgression, whereas comparative phylogenomic frameworks and diversification models can test whether introgression systematically promotes speciation and potentially fuels evolutionary radiations (e.g., Meier et al. 2017; Svardal et al. 2020; Qian et al. 2023; Yardeni et al. 2025). Such work is especially important in a climate context, since at a macroevolutionary scale, the role of climate heterogeneity in speciation has been well documented across numerous plant and animal taxa, with generally higher diversification rates observed in warm periods or regions (e.g., Mittelbach et al. 2007; Quintero and Jetz 2018; Rangel et al. 2018; Condamine et al. 2019; Juvé et al. 2025).

5.2 | Genomic Offset, Adaptive Introgression and Rescue

Genomic offset is an emerging approach for predicting future maladaptation under rapid environmental change, especially climate change (Fitzpatrick and Keller 2015; Rellstab et al. 2021). The rationale of these methods is to estimate the potential mismatch between current and future environmental conditions, through the distance between the current and future optimal genetic composition inferred from genotype-environment associations. However, genomic offset approaches are inherently limited to standing genetic variation that is both present in the focal population and statistically associated with contemporary environmental gradients. Thus, these approaches cannot account for adaptive alleles that are currently absent or rare including those that could be introgressed from related taxa in the future. Nevertheless, comparing genomic offset maps across closely related species could, in principle, help identify populations harbouring potentially beneficial alleles and predict directions for future adaptive introgression. A similar approach has been used to predict source and sink populations for evolutionary rescue potential in the bat species *Myotis escalerai* and *M. crypticus* (Razgour et al. 2019), and also to predict sources and destinations of future potential invasions (Camus et al. 2024). In this context, if one taxon shows a lower genomic offset under projected future conditions, it could act as a potential donor species and serve as a genetic rescuer. If and when such gene flow

will indeed happen, either naturally or through human-assisted movement, these alleles are expected to introgress, reducing their genomic offset of the recipient species and thus providing a genomic rescue. As a result, combining offset-based vulnerability assessments with introgression analyses could provide a powerful framework for forecasting adaptive responses in changing environments.

5.3 | Community-Level Investigations

Investigating adaptive introgression has important consequences for conservation and management in the face of climate change and land-use change. Recognising this role does not mean adopting a simplistic view of adaptation. Adaptive introgression studies generally focus on a single focal species, whereas the fitness of this species is often associated with a wide community of organisms, from closely associated species (gut microbiota, plant phyllosphere, etc.) to more distant interactions in complex ecosystems. Although estimating the precise contribution of adaptive introgression to adaptation is currently an important step forward, current research mainly targets single species. It is now time to explore how entire communities of symbiotic organisms—the so-called super-organisms—adapt over short timescales, and to better understand the true contribution of adaptive introgression at a broader scale. For many organisms, sequencing whole individuals (e.g., insects) or specific organs (e.g., gut, leaves) can provide valuable information on microbial diversity through metagenomic approaches. In fact, most studies focused on a single species begin by mapping sequencing reads to a single reference genome, discarding all other DNA. Generating genomic resources (metassembling) allows investigating how these other species adapt, including the microbiome itself. Environmental DNA (eDNA) approaches are expected to become a key future direction in order to estimate the contribution of adaptive introgression at the level of community genomes.

Author Contributions

T.L. wrote the Sections 1, 2, 3 and 5. M.H. carried out the literature search related to adaptive introgression in a climate context and wrote the associated Section 4. Both authors contributed to the revision and approval of the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The authors have nothing to report.

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